

***Suggested Teaching Guidelines for
Natural Language Processing & Computer Vision
PGCP-AI February 2026***

Duration: 40 hours Theory + 60 hours Lab + 20 Hours Self-Learning

Objective: To Introduce the students to Natural Language Processing & Computer Vision.

Prerequisites: Good knowledge of Python Programming and fundamentals of Artificial Intelligence.

Evaluation method:

Theory exam–	40%
Lab Exam -	40%
Internal exam-	20%

List of Books / Other training material

Courseware:

Natural Language Processing, by Pushpak Bhattacharyya, Aditya Joshi / Wiley India, 2023

Reference Book:

1. The Big Book of NLP Techniques by Shlomo Vaknin
2. Theory and Practice of NLP Coaching: A Psychological Approach by Bruce Grimley

NLP (26 Hrs Theory + 44 Hrs Lab + 15 Hrs Self Learning)

Session 1: (2T + 2L + 1SL)

Theory and Lab

Language in Cognitive Science

- Definitions of language
- Language as a rule-governed dynamic system
- Knowledge of language: Innateness of Grammar
- Language as a biological, social and psychological phenomenon
- Modes of language: spoken and written
- Language system as expression and content
- Language and symbolic systems: Artificial language (Logical language/programming Language) vs. Natural Language
- Linguistics as a scientific study

Self-Learning:

- Explore how language is modeled in AI systems (chatbots, voice assistants) and observe how rules vs learning differ in Grammarly vs ChatGPT
- Reflect why did rule-based NLP fail at scale and how does human ambiguity affect AI systems
- Compare Natural Language vs Programming Language in AI systems and study how symbolic AI evolved into statistical NLP

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Session 2: (2T + 2L + 1SL)**Theory and Lab****Language Analysis and Computational Linguistics**

Language Analysis:

- Paradigmatic and Syntagmatic relationship
- Form, Function and Meaning in Language Analysis
- Levels of Linguistic Analysis: Phonetics, Phonology, Morphology, Syntax, Semantics,
- Discourse, Pragmatics, Lexicology
- Artificial Intelligence (AI) and its sub disciplines
- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)
- Natural Language Interaction (NLI)

Assignment: Difference between Semantics and Pragmatics

Self-Learning:

- Observe differences between semantic correctness and pragmatic correctness
- Reflect why AI sometimes understands meaning but fail context and where does pragmatics, break NLP systems
- Read about linguistic levels used in NLP pipelines
- Prepare a concept map linking: Syntax → Semantics → Pragmatics → Applications

Session 3: (2T+ 4L+ 1 SL)**Theory****Shallow Parsing and Tools for NLP**

- Morphological Analysis
- Tokenization & PoS Tagging
- Chunking & Multi word expression (MWE)
- Named-Entity Recognition
- Lemmatizer & Stemming
- Morphological Synthesis
- Word Sense Disambiguation
- Universal Networking Language

Lab Assignment: Apply tok , stemming , lemma, and POS tagging for following data <https://www.kaggle.com/arushchillar/disneyland-reviews>

Self-Learning:

- Explore and compare NLTK vs spaCy vs HuggingFace tokenizers
- Visualize POS tagging and NER outputs
- Study and understand why tokenization differs across languages and why stemming fails for semantic tasks

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Session 4: (2T+ 4L + 1SL)

Theory and Lab

Deep Parsing and Tools for NLP

- Syntactic Parsing Techniques and algorithms
- Semantic Parsing
- Information Extraction
- Automatic Summarization
- Anaphora Resolution, Pragmatics and Discourse analysis
- Ontology & Semantic Web

Self-Learning:

- Explore and try dependency parsing visualization (spaCy displaCy)
- Understand why is semantic parsing still hard and why do summaries hallucinate
- Research knowledge graphs and Understand ontologies in healthcare / finance NLP

Session 5: (2T+ 4L + 1SL)

Theory and Lab

Statistical Approaches

- Probability Theory & Models
- Discrete Time Models
- Markov Models, Entropy Models
- Statistical Parsing
- Text Categorization / Classification and Clustering
- Text Classification Using Support Vector Machine (SVM)
- Centroid based Classification

Self-Learning:

- Explore spam classifiers and news categorization systems and Observe misclassifications
- Understand and study why probabilistic models still matter in DL era
- Explore Trade-off: interpretability vs performance.
- Research and Study Markov Models vs Neural Models, understand entropy in language modeling

Session 6: (2T+ 4L+ 2SL)

Theory and Lab

NLP with Machine Learning and Deep Learning

- Machine Learning & Deep Learning for NLP
- ML vs. DL along with respective algorithms names
- Linear Algebra Review
- ANN, RNN, CNN, DNN Long Short-Term Memory (LSTM)
- Gated Feedback Recurrent Neural Networks

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Assignment: Difference between ANN, RNN and CNN

Self-Learning:

- Explore use cases about ANN vs RNN vs CNN in NLP
- Observe how RNN remembers sequence and study gradient problems in deep learning
- Try and Implement comparison table + demo notebook using ANN vs RNN vs CNN on dataset

Session 7: (2T+ 4L+ 2SL)

Theory and Lab

Study of Important ML/DL Algorithms & Case Studies:

- Pre-processing
- Need of Pre-processing Data
- Introduction to NLTK, spaCy
- Using Python Scripts

Self-Learning:

- Explore spaCy pipelines, custom components and analyze pre-processing effects
- Understand and study is preprocessing always helpful and When raw text beats cleaned text
- Research industry NLP pipelines (search engines, recommender systems)
- Build end-to-end NLP pipeline: Raw text → clean → vectorize → classify

Session 8: (2T + 4L + 1SL)

Theory

- Introduction to gates, GRU and LSTM
- How LSTM and GRU resolves exploding and vanishing gradient problems

Assignment –Lab:

- Implement LSTM and GRU in python
- Sequence Modeling Lab
- Using Tensor Flow and python for music
- Generation using deep learning

Self-Learning:

- Understand and analyse gate activations in LSTM and explore sequence generation demos
- Understand and study why LSTM memory cell matters, why GRU is faster
- Research LSTM used in Speech recognition and Time-series forecasting

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Session 9: (2T + 4L + 1SL)**Theory and Lab**

- Sequence Modeling - Word representation
- word embedding matrix
- learning words and embedding's
- Sentiment classifications

Self-Learning:

- Explore embeddings using t-SNE / PCA and Observe semantic similarity
- Understand and study why embeddings capture bias, Static vs contextual embeddings
- Train embeddings on custom dataset and perform sentiment classification using embeddings

Session 10: (2T + 4L + 2SL)**Theory and Lab****Word2Vec models (Skip-gram, CBOW, Glove, one hot Encoding)**

- Sequence-to-sequence models (Seq2Seq) - GloVe: Global Vectors for Word representation
- Brief on Sequence-to-Sequence Models
- RNN based Sequence-to-Sequence Model
- Challenges
- Transformer in NLP
- Understanding the Model Architecture
 - Getting Hang of Self- Attention
 - Calculation of Self-Attention
 - Limitations of the Transformer
- Transformer-XL oUsing Transformer for Language Modeling
- Using Transformer- XL for Language Modeling

Self-Learning:

- Explore and Use HuggingFace Transformers for inference
- Visualize self-attention weights
- Study why attention replaces recurrence and Transformer scalability challenges
- Research Transformer-XL vs vanilla Transformer

Session 11: (2T + 4L + 1SL)**Theory and Lab**

- Bert in NLP
- Model Architecture
- BERT Pre-Training Tasks
- NLP Model Deployment Techniques using Flask

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Self-Learning:

- Explore and study BERT embeddings for classification
- Observe and document difference from Word2Vec
- Understand and study why pre-training matters and When fine-tuning fails
- Learn REST APIs for ML models
- Deploy NLP model using Flask and Test with Postman

Session 12 & 13: (4T + 4L + 1SL)**Theory****Speech Processing**

- Articulatory Phonetics
- Speech Sounds and Phonetic Transcription
- Acoustic Phonetics
- Phonology
- Computational Phonology
- Digital Signal Processing Techniques
- Automatic Speech Recognition (ASR)
- Speech Recognition Approaches
- Text to Speech (TTS) system
- Speech Synthesis Approaches
- Language Models

Lab**NLP Applications**

- Lexicon, Dictionaries, thesaurus
- Transliteration, Spell Checker, Grammar Checker, Domain identification, Language identification
- Auto suggest/ Auto complete, Gender Prediction
- Machine Translation
- Neural machine translation
- Information extraction and Retrieval
- Question answering & dialogue agents
- Speech Technologies
- OCR, Hand Writing Recognition
- Indian Language Script Technology
- Language identification
- Auto suggest/ Auto complete, chat bots, Robotics

Self-Learning:

- Explore speech-to-text APIs
- Analyze waveform and spectrograms
- Understand and study why accents affect ASR?
- Study ASR pipelines and explore Indian language speech challenges

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Computer Vision (14 Hrs Theory + 16 Hrs Lab + 5 Hrs Self-Learning)

Session 14 & 15: (4T + 4L + 1SL)

Theory

- Introduction to Computer Vision
- Computer Vision and Natural Language Processing
- The Three R's of Computer Vision
- Basics of Image Processing
- Vision Level – Low, Mid & High

Lab Assignments: Load any .jpeg image and process the image using Open CV

Self-Learning:

- Explore and Visualize image channels and pixel values.
- Try OpenCV transformations.
- Study and understand why CV ≠ image processing and why CV + NLP is powerful
- Build image preprocessing pipeline

Session 16 & 17: (4T + 4L + 1SL)

- Edge Detection, Interest Points and Corners
- Image Classification
- Recognition, Bag of Features, and Large-scale Instance Recognition
- Transfer Learning

Lab Assignments: Load any image and classify the image and implement the edge detection

Self-Learning:

- Compare traditional features vs CNN features
- Understand and study why handcrafted features failed
- Research transfer learning in industry

Session 18: (2T + 4 L + 1SL)

- Alex Net,
- ResNet,
- Image Net

Lab Assignments: Implement AlexNet, ResNet and Image Net on the data sets.

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Self-Learning:

- Explore and Compare AlexNet vs ResNet depth
- Understand and study why residual connections work
- Research study ImageNet's impact

Session 19 & 20: (4T + 4L + 2SL)

Introduction to Object Detection Algorithms

- YoLo (High Level Overview)
- RCNN
- Fast RCNN
- Faster RCNN
- Mask RCNN

Lab Assignments:

Applications of Computer Vision

- Gender Prediction
- Image Mapping
- Face / Object Recognition
- Image Recreation
- Image Matching

Self-Learning:

- Research about Study CV use cases:
- Surveillance
- Retail analytics
- Healthcare imaging
- Study how Agentic AI systems compare to traditional rule-based and reactive AI models- Explore autonomous decision-making and how agents select actions under uncertainty
- Study reflective self-correction and how feedback loops help agents revise their behavior
- Analyze context-aware behavior and the role of memory, goals, and environment in agent decision-making
- Explore continuous learning architectures that allow agents to adapt over time rather than remain static